



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.NOS.8

Compare and Order Integers and Rational Numbers

Lesson 7-4 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can compare and order integers using a number line.

Language Objective: I can explain my reasoning using the words compare, order, greater than, and less than.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK – FILL EACH BLANK WITH THE BEST WORD

Number line

Compare

Order

Greater than

Less than

Negative integer

Integer

Inequality



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

To plot integers in order, place them on a .

2

To decide which number is larger or smaller is to them.

3

To arrange numbers from least to greatest or greatest to least is to them.

4

On a number line, a number farther to the right is another number.

5

On a number line, a number farther to the left is another number.

6 An integer less than zero, like -6 , is a .

7 A whole number or its opposite is an .

8 A statement that compares numbers with $<$ or $>$ is an
 .

Write About the Math

How do you use integer comparison to find the coldest and warmest days?

¿Cómo usas la comparación de enteros para hallar el día más frío y el más caluroso?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Number line**
Recta numérica

☐ **Compare**
Comparar

☐ **Order**
Ordenar

☐ **Greater than**
Mayor que

☐ **Less than**
Menor que

Model: *-3 is less than 2 because on the number line ____.* — Student explains that -3 is farther left on the number line than 2, and numbers decrease moving left.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The coldest day was ____ at ____°F because ____.** **The warmest day was ____ at ____°F. I ordered them from coldest to warmest by ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Compare
3. **Notes 3:** Order
4. **Notes 4:** Greater than
5. **Notes 5:** Less than
6. **Notes 6:** Negative integer
7. **Notes 7:** Integer
8. **Notes 8:** Inequality
9. **Watch for this mistake:** *A common mistake in Compare and Order Integers is thinking that a negative number with a bigger digit is the bigger value — for example, believing -15 is greater than -8 because 15 is greater than 8. It's the opposite for negatives: the number farther to the left on the number line (farther from zero) is always the LESSER value, so $-15 < -8$. Before you submit, check your comparison against the number line, not just the digits.*
10. **Try It 1:** D. -4 m — *Closer to zero means greater for negatives. -4 is only 4 below the surface, so -4 is the greatest of -9, -4, and -15.*
11. **Try It 2:** D. $<$ — *-8 is farther left (deeper) than -3, so -8 is less than -3. The symbol is $<$.*